

JASMIKA V

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building cloud-native applications, Python backend services, REST APIs, and AI-enabled solutions on AWS. Experienced in FastAPI, Java, event-driven architectures, Generative AI, Amazon Bedrock, LangChain, and GraphQL. Proven track record of developing scalable backend systems, integrating ML inference into production applications, and delivering enterprise solutions for aviation, government, and healthcare domains. IEEE-published NLP researcher with a Master's in Computer Science from Georgia State University.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash, C#

AI/ML: Amazon SageMaker, AWS Bedrock, LangChain, XGBoost, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NLP, Computer Vision, RAG, Generative AI, Vector Databases (Pinecone, FAISS)

Cloud & DevOps: AWS Lambda, Step Functions, EventBridge, API Gateway, ECS Fargate, DynamoDB, S3, CloudWatch, SNS, Cognito, WAF, GitLab CI, Terraform, Docker, Kubernetes

Frameworks & APIs: FastAPI, Django REST Framework, GraphQL, REST APIs, React, Angular, PySpark

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB

Tools & Practices: Git, JIRA, Postman, Swagger, Amazon Q, Kiro, pytest, Agile/Scrum, CI/CD

WORK EXPERIENCE

Delta Air Lines, Franklin Infotech, Full Stack Developer, Atlanta

August 2024 - Present

- Architected **cloud-native** backend services and REST APIs for the CORE and DGT platforms using Python, FastAPI, AWS Lambda, EventBridge, DynamoDB, and SageMaker, integrating ML inference into event-driven operational recovery workflows.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline using **AWS Bedrock** and LangChain to generate business-friendly explanations for recovery recommendations, shortening analyst investigation time by 40%.
- Engineered event-driven workflows to process real-time flight-change events within a 36-hour operational recovery window, enabling timely prediction requests and downstream updates.
- Integrated **SageMaker** inference endpoints with AWS Lambda, DynamoDB, and S3 to process over 1,000 daily flight-change events and persist prediction results for downstream applications.
- Established a **GitLab CI MLOps framework** with automated drift detection and regression gates, cutting model release time by 50% while promoting models across dev, staging, and prod.
- Orchestrated **CORE ETL workflows** with AWS **EventBridge**, **Step Functions**, and **PySpark** transformations, loading aircraft schedules and crew assignments into DynamoDB and reducing processing errors by 25%.
- Developed a **GraphQL** interface using Python **FastAPI** on AWS Lambda to serve model-ready datasets to the CORE backend solver, cutting solver data-fetch time by 20%.
- Led **Angular** UI development for CORE recovery dashboards (Fleet, Crew Cancel, Station Capacity) using RxJS and NgRx, decreasing manual scheduling effort by 40%.
- Implemented **CloudWatch** monitoring and SNS alerts to improve application observability, accelerating incident detection by 45%.
- Collaborated with operations and data science teams to deliver AI-enabled recovery features while leveraging **Amazon Q** and **Kiro** to improve development productivity.
- Containerized backend services using Docker to standardize deployments across development, testing, and production environments.

Georgia State University - GTA, AWS Cloud-Native Developer, Atlanta

August 2023 - June 2024

- Mapped legacy **webMethods ESB** functionality to a cloud-native migration strategy, scoping cutover criteria across 12+ partner agency integrations that became the blueprint for a phased, zero-downtime rollout.
- Executed migration of the **SOAP ESB** to an AWS cloud-native architecture, completing cutover for all state agency consumers with zero downtime.

- Replaced legacy **SOAP** interfaces with a reusable microservices framework of Python RESTful services, cutting new agency integration onboarding from 3 weeks to 2 days.
- Hardened the API surface with AWS **WAF**, Mutual TLS, and a custom RBAC model using OAuth2 M2M authorization via AWS Cognito, enforcing least-privilege access across 15+ client agencies.
- Created a **React** portal with a hooks-based component library, giving agency teams self-service visibility into onboarding status and service health.
- Optimized compute spend by shifting non-critical workloads to AWS ECS Spot Fargate, lowering infrastructure costs by 35%.
- Provisioned **Infrastructure as Code** with **Terraform** and AWS CodePipeline, standardizing repeatable deployments and eliminating configuration drift across all environments.
- Configured **API Gateway** endpoints with usage plans, throttling, and request validation, standardizing secure external access across all partner agency integrations.

Georgia State University, Software Developer, Atlanta

August 2022 - May 2023

- Deployed **FastAPI** middleware APIs serving medical device analytics to 5+ downstream healthcare systems, with authentication, input validation, and structured error handling.
- Migrated REST endpoints to **GraphQL** using Graphene and Ariadne, enabling client-driven queries and narrowing average payload size by 40%.
- Delivered **WebDocAnalyzer**, an NLP pipeline using TF-IDF, NER, and Graph-based HITS algorithms, achieving 77.9% precision on a news corpus.
- Crafted a responsive **React UI** with functional components and Context API, letting clinicians filter and export device telemetry across 20+ views, and authored pytest suites that lifted code coverage to 85%.

Yinbotech Systems, Software Associate

January 2022 - July 2022

- Analyzed business and technical requirements to design scalable Python backend services, then built **ETL pipelines** processing 100K+ client records daily into MySQL with Regex-based validation to catch malformed data before downstream systems.
- Extended RESTful APIs with **Django REST Framework**, adding custom role-based authorization middleware across 10+ endpoints, and designed optimized relational database schemas with SQL queries and stored procedures to maintain data integrity and performance.
- Enforced authentication, authorization, and data validation logic, then validated auth flows and API contracts using Python **unittest** and DRF's test client, fixing 15+ bugs before QA handoff.
- Conducted peer code reviews and standardized reusable Python utility modules for logging, error handling, and validation, which were reused across 3+ teams to maintain consistent code quality.

WOIR Software India Pvt. Ltd., Software Developer Intern

April 2021 - August 2021

- Co-developed a **Computer Vision color recommendation system** using OpenCV-Python and K-means clustering, generating ML-driven complementary color suggestions with 80%+ user-rated relevance for garments.

Indian Servers India, Web Developer & 3D Designer Intern

April 2020 - August 2020

- **Modeled 15+ 3D game assets** in Unity with C# and integrated Unity WebGL builds for web platforms, enabling browser-based deployment across 4+ major browsers.

EDUCATION

Georgia State University - Master of Science in Computer Science, Atlanta, GA

PVP Siddhartha Institute of Technology - Bachelor of Technology in Computer Science and Engineering, India

CERTIFICATIONS AND PUBLICATIONS

AWS Certified Cloud Practitioner - Amazon Web Services.

IEEE Publication: "[News Classification Using NLP](#)" - ICIEM 2022.